

We now turn to a very important notion that you may not have met before.

1.3.7 Definition. Let f be a function from a set X into a set Y . If S is any subset of Y , then the set $f^{-1}(S)$ is defined by

$$f^{-1}(S) = \{x : x \in X \text{ and } f(x) \in S\}.$$

The subset $f^{-1}(S)$ of X is said to be the **inverse image** of S .

Note that an inverse function of $f: X \rightarrow Y$ exists if and only if f is bijective. But the inverse image of any subset of Y exists even if f is neither one-to-one nor onto. The next example demonstrates this.

1.3.8 Example. Let f be the function from the set of integers, $\mathbb{Z}$, into itself given by $f(z) = |z|$, for each $z \in \mathbb{Z}$.

The function f is not one-to one, since $f(1) = f(-1)$.

It is also not onto, since there is no $z \in \mathbb{Z}$, such that $f(z) = -1$. So f is certainly not bijective. Hence, by [Proposition 1.3.6 \(i\)](#), f does not have an inverse function. However inverse images certainly exist. For example,

$$f^{-1}(\{1, 2, 3\}) = \{-1, -2, -3, 1, 2, 3\}$$

$$f^{-1}(\{-5, 3, 5, 7, 9\}) = \{-3, -5, -7, -9, 3, 5, 7, 9\}.$$

□

We conclude this section with an interesting example.

1.3.9 Example. Let $(Y, \mathcal{T})$ be a topological space and X a non-empty set. Further, let f be a function from X into Y . Put $\mathcal{T}_1 = \{f^{-1}(S) : S \in \mathcal{T}\}$. Prove that $\mathcal{T}_1$ is a topology on X .

Proof.

Our task is to show that the collection of sets, $\mathcal{T}_1$, is a topology on X ; that is, we have to show that $\mathcal{T}_1$ satisfies conditions (i), (ii) and (iii) of [Definitions 1.1.1](#)

$$\begin{aligned} X \in \mathcal{T}_1 & \text{ since } X = f^{-1}(Y) \text{ and } Y \in \mathcal{T}. \\ \emptyset \in \mathcal{T}_1 & \text{ since } \emptyset = f^{-1}(\emptyset) \text{ and } \emptyset \in \mathcal{T}. \end{aligned}$$

Therefore $\mathcal{T}_1$ has property (i) of [Definitions 1.1.1](#).

To verify condition (ii) of [Definitions 1.1.1](#), let $\{A_j : j \in J\}$ be a collection of members of $\mathcal{T}_1$, for some index set J . We have to show that $\bigcup_{j \in J} A_j \in \mathcal{T}_1$.

As $A_j \in \mathcal{T}_1$, the definition of $\mathcal{T}_1$ implies that $A_j = f^{-1}(B_j)$, where $B_j \in \mathcal{T}$. Also $\bigcup_{j \in J} A_j = \bigcup_{j \in J} f^{-1}(B_j) = f^{-1}\left(\bigcup_{j \in J} B_j\right)$. [See [Exercises 1.3 # 1](#).]

Now $B_j \in \mathcal{T}$, for all $j \in J$, and so $\bigcup_{j \in J} B_j \in \mathcal{T}$, since $\mathcal{T}$ is a topology on Y . Therefore, by the definition of $\mathcal{T}_1$, $f^{-1}\left(\bigcup_{j \in J} B_j\right) \in \mathcal{T}_1$; that is, $\bigcup_{j \in J} A_j \in \mathcal{T}_1$.

So $\mathcal{T}_1$ has property (ii) of [Definitions 1.1.1](#).

[Warning.] You are reminded that not all sets are countable. (See the Appendix for comments on countable sets.) So it would not suffice, in the above argument, to assume that sets $A_1, A_2, \dots, A_n, \dots$ are in $\mathcal{T}_1$ and show that their union $A_1 \cup A_2 \cup \dots \cup A_n \cup \dots$ is in $\mathcal{T}_1$. This would prove only that the union of a countable number of sets in $\mathcal{T}_1$ lies in $\mathcal{T}_1$, but would not show that $\mathcal{T}_1$ has property (ii) of [Definitions 1.1.1](#)— this property requires all unions, whether countable or uncountable, of sets in $\mathcal{T}_1$ to be in $\mathcal{T}_1$.]

Finally, let A_1 and A_2 be in $\mathcal{T}_1$. We have to show that $A_1 \cap A_2 \in \mathcal{T}_1$.

As $A_1, A_2 \in \mathcal{T}_1$, $A_1 = f^{-1}(B_1)$ and $A_2 = f^{-1}(B_2)$, where $B_1, B_2 \in \mathcal{T}$.

$$A_1 \cap A_2 = f^{-1}(B_1) \cap f^{-1}(B_2) = f^{-1}(B_1 \cap B_2). \quad [\text{See } \textcolor{blue}{\text{Exercises 1.3 \#1.}}]$$

As $B_1 \cap B_2 \in \mathcal{T}$, we have $f^{-1}(B_1 \cap B_2) \in \mathcal{T}_1$. Hence $A_1 \cap A_2 \in \mathcal{T}_1$, and we have shown that $\mathcal{T}_1$ also has property (iii) of [Definitions 1.1.1](#).

So $\mathcal{T}_1$ is indeed a topology on X . □

Exercises 1.3

1. Let f be a function from a set X into a set Y . Then we stated in [Example 1.3.9](#) that

$$f^{-1}\left(\bigcup_{j \in J} B_j\right) = \bigcup_{j \in J} f^{-1}(B_j) \quad (1)$$

and

$$f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2) \quad (2)$$

for any subsets B_j of Y , and any index set J .

- (a) Prove that (1) is true.

[Hint. Start your proof by letting x be any element of the set on the left-hand side and show that it is in the set on the right-hand side. Then do the reverse.]

- (b) Prove that (2) is true.

- (c) Find (concrete) sets A_1, A_2, X , and Y and a function $f: X \rightarrow Y$ such that $f(A_1 \cap A_2) \neq f(A_1) \cap f(A_2)$, where $A_1 \subseteq X$ and $A_2 \subseteq X$.

2. Is the topology $\mathcal{T}$ described in Exercises 1.1 #6 (ii) the finite-closed topology? (Justify your answer.)

T_1 -spaces

3. A topological space $(X, \mathcal{T})$ is said to be a **T_1 -space** if every singleton set $\{x\}$ is closed in $(X, \mathcal{T})$. Show that precisely two of the following nine topological spaces are T_1 -spaces. (Justify your answer.)

- (i) a discrete space;
- (ii) an indiscrete space with at least two points;
- (iii) an infinite set with the finite-closed topology;
- (iv) [Example 1.1.2](#);
- (v) [Exercises 1.1 #5 \(i\)](#);
- (vi) [Exercises 1.1 #5 \(ii\)](#);
- (vii) [Exercises 1.1 #5 \(iii\)](#);
- (viii) [Exercises 1.1 #6 \(i\)](#);
- (ix) [Exercises 1.1 #6 \(ii\)](#).

4. Let $\mathcal{T}$ be the finite-closed topology on a set X . If $\mathcal{T}$ is also the discrete topology, prove that the set X is finite.

T_0 -spaces and the Sierpinski Space

5. A topological space $(X, \mathcal{T})$ is said to be a **T_0 -space** if for each pair of distinct points a, b in X , either² there exists an open set containing a and not b , or there exists an open set containing b and not a .
- (i) Prove that every T_1 -space is a T_0 -space.
- (ii) Which of (i)–(vi) in Exercise 3 above are T_0 -spaces? (Justify your answer.)
- (iii) Put a topology $\mathcal{T}$ on the set $X = \{0, 1\}$ so that $(X, \mathcal{T})$ will be a T_0 -space but not a T_1 -space. [The topological space you obtain is called the **Sierpiński space**.]
- (iv) Prove that each of the topological spaces described in Exercises 1.1 #6 is a T_0 -space. (Observe that in Exercise 3 above we saw that neither is a T_1 -space.)

Countable-Closed Topology

6. Let X be any infinite set. The **countable-closed topology** is defined to be the topology having as its closed sets X and all countable subsets of X . Prove that this is indeed a topology on X .

²You are reminded that the use of “or” in mathematics is different from that in everyday English. In mathematics, “or” is not exclusive. See the comments on this in Chapter 0.